AZURE / KUBERNETES / TERRAFORM

One app.
Two AKS replicas.

A smaller stack, with app identity instead of a storage secret.

Browser → Azure LoadBalancer (HTTP)



AKS / TWO-NODE POOL

Node A

Python app replica

Same immutable image

Node B

Python app replica

Same immutable image



Workload identity

Private blob container: app-data

ACR stores one image; deployment pins its digest.

Terraform provisions 10 resources in one root configuration.

Evidence, with its limits.

Recorded live check / November 1, 2026 / westus3

Health requests

100

10 concurrent clients

Request failures

0

In the final public sample

Median latency

306.7 ms

Public client-to-app path

p95 latency

448.1 ms

Public client-to-app path

6 blob upload / read / delete cycles passed

Two Ready replicas on separate nodes; zero restarts.

First-run p95: 3825.9 ms (also retained).

Short samples. No production capacity or uptime claim.
Source: evidence/live-2026-11-01/*.json

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A green rollout hid a placement bug.

Testing the final state mattered as much as rollout success.

OBSERVED AFTER THE FIRST RESTART

Node A	Node B
0 new replicas	**2 new replicas**

Fix: spread each revision independently

matchLabelKeys: [pod-template-hash]

VERIFIED AFTER ANOTHER RESTART

Node A	Node B
1 Ready replica	**1 Ready replica**

28 health checks; 0 observed request failures.

Recorded outputs: rollout.json and workloads.txt
Illustration of recorded state; not a dashboard screenshot.

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Clone it. Run it. Check the evidence.

Setup, troubleshooting and cleanup are included in GitHub.

01 Run locally

Clone → virtual environment → seven local tests.

02 Provision Azure

Choose your subscription → backend → Terraform plan/apply.

03 Deploy and verify

Build one image → digest-pinned rollout → HTTP/blob checks.

04 Inspect and clean up

Review raw results → remove app resources when finished.

Deliberately a PoC, not production-ready.

HTTP • no zone redundancy • quota headroom needed

Azure resources incur charges. No proven node-failure recovery.